SWEDISH MANUFACTURING CONFERENCE · 20–21 MAY 2026

DYNAMIC SALSA

Warehouses that are flexible.

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THE HIDDEN COST

20%

of all logistics cost is just warehousing — space, labour, and the chaos of change.



MEET THE PICKER

It all rides on the individual.

High mix

Manual

Every operator differs

Fewer, newer people

WHY WAREHOUSES STAY STUCK

We're running half-blind.

Gap 1 · Information

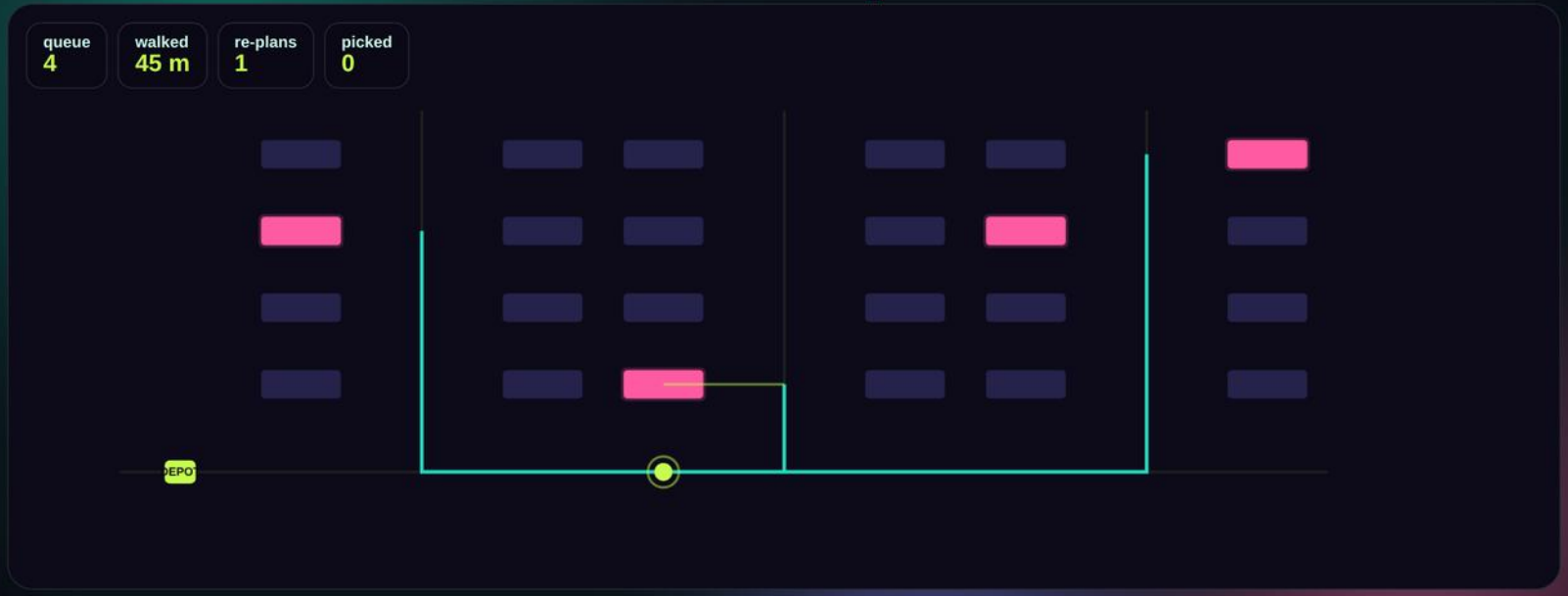
Not enough information — we rarely know what operators really do, or how long it takes.

Gap 2 · Workload

Unbalanced, unsustainable workload — heavy items, repetition and one-size-fits-all routes, rarely tuned to the individual.

WHAT “DYNAMIC” LOOKS LIKE

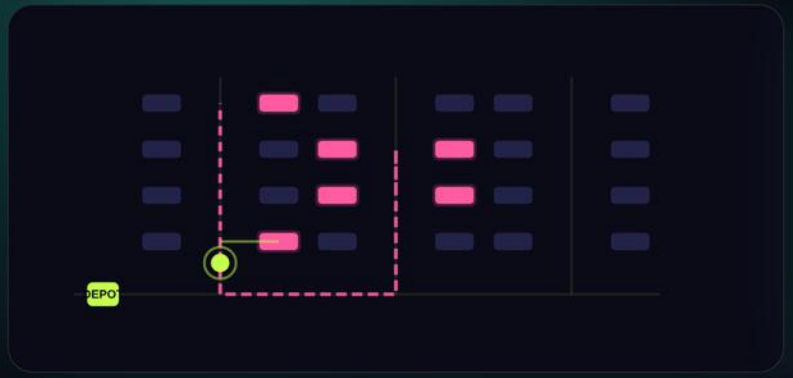
Watch the route re-plan itself.



THE WHOLE IDEA, IN ONE PICTURE

A bus line — or on-demand?

BUS fixed route



Bus — walked 36 m

ON-DEMAND goes where needed



On-demand — walked 36 m

On-demand picking = dynamic scheduling.

HOW WE WORKED

We started with people.

Watched operators

Mapped real tasks

Designed around them

AI for resilience under change
— privacy first.



THE OPERATOR'S FOUR QUESTIONS

One workflow, four answers.

01 **Where to go?**
AI schedules the picking route around demand, distance and workload.

02 **What to do?**
The right info on a tablet, at the moment of the pick.

03 **How did we do?**
A camera reads what happened — setup, search, travel, picking.

04 **How to improve?**
Data feeds back; the schedule keeps learning.

FROM THE FACTORY TO THE LAB

A real problem, brought home.



The real problem
Order picking at Scania

→



Designed together
Data, software, partners

→

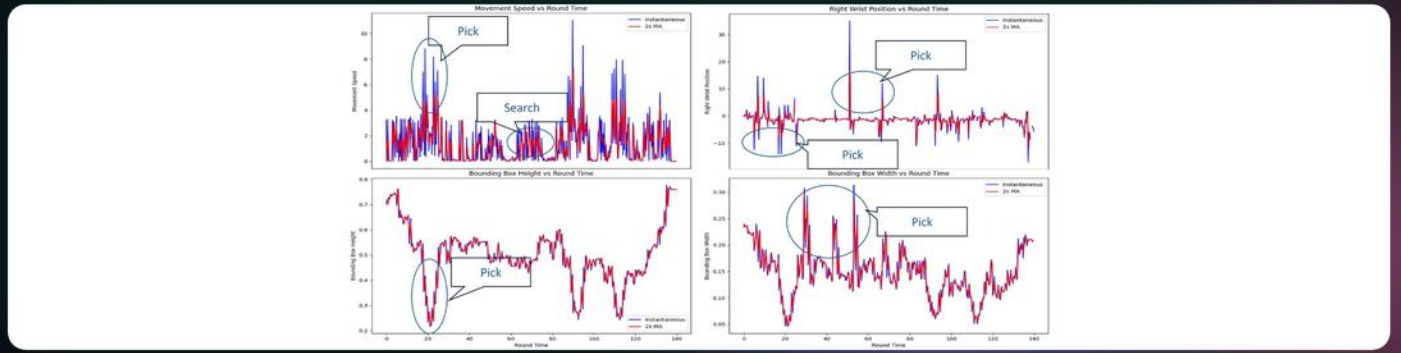


Prototyped & tested
KTH Production Logistics Lab

HOW DID WE DO?

We taught a camera to read work.





THE SURPRISE

Picking is barely a tenth of it.



Travel 44% Search 23% Setup 22% Picking 11%

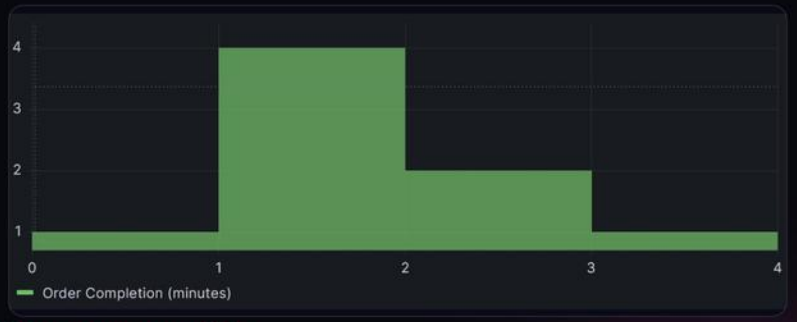
Travel, search and setup eat the day — exactly what smarter scheduling can cut.

THE KPIS WE FEED BACK

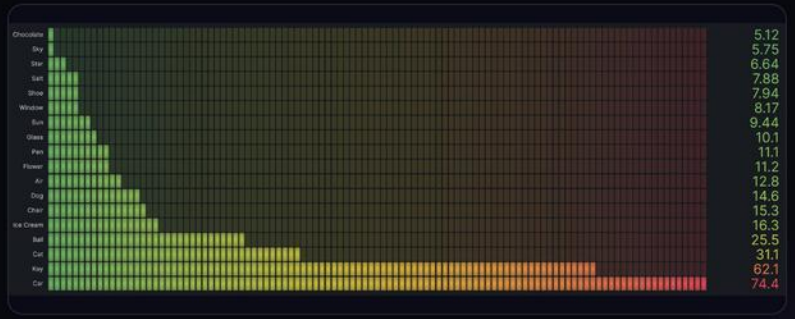
More visibility, closer to real time.



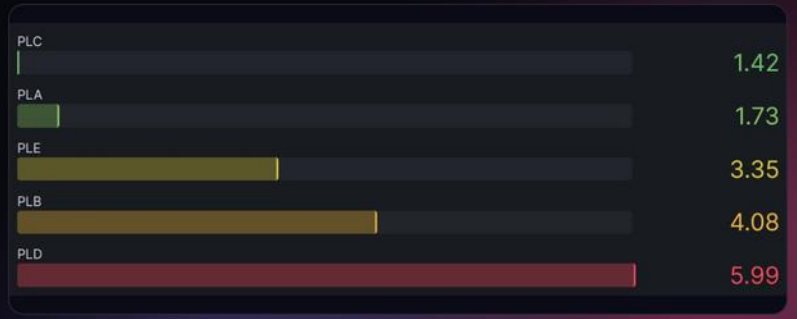
Mean lead time · picking list



Frequency vs. completion



Picking time per product



Order completion time

THE PLOT TWIST

The same camera reads the operator.



It turns motion into standard time — even per individual — plus a read on ergonomic load. (Our TIMEBLY 2 project, Vinnova.)

movavi

CAMERAS ARE SENSITIVE

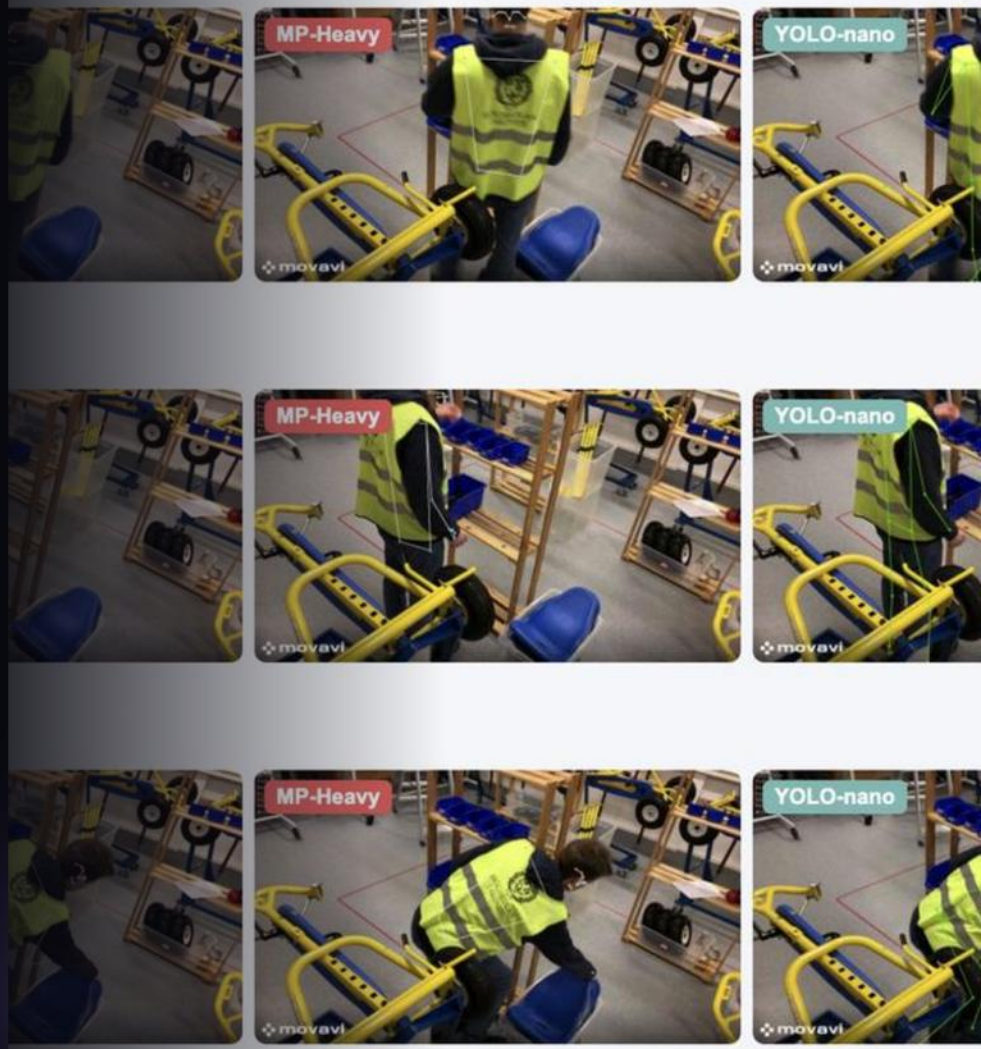
Skeletons, not faces.

Skeleton only

Faces blurred

Support, not surveillance

Better, safer work — not watching people.



BACK TO WHERE WE STARTED

The two gaps — narrower.

Gap 1 · Information

Not enough information → **improved with AI vision.** More of the work becomes visible, closer to real time.

Gap 2 · Workload

Unbalanced workload → **improved with workload-aware scheduling.** Balances operator workload & cart capacity, not just distance (Wang et al., 2025).

BENEFITS & IMPACT

Who wins — and how.

Who benefits

Operators & engineers · companies big & small · standardisation bodies · universities & students.

Impacts

Well-being & performance · privacy & data protection · gender equality & diversity · the value of human work.

Markets

New products & digital services · cognitive support · resilience · flexibility.

PART OF A BIGGER TEAM

A Sweden–Korea project.

Sweden · logistics

Our side: order picking with AI vision & adaptive scheduling.

Korea · assembly

The same vision idea on railway-vehicle manufacturing — object tracking & a digital twin.

Funded through Eureka SMART — two consortia, one idea.

NOT AN ENDING — A DIRECTION

Where this is heading.

Human / AI Collaboration for the Workforce of the Future

Horizon Europe · HORIZON-CL4-2027 — co-learning, adaptive automation, well-being & inclusion.

Still at the intention stage — if it resonates,
come talk to us.

LET'S TALK

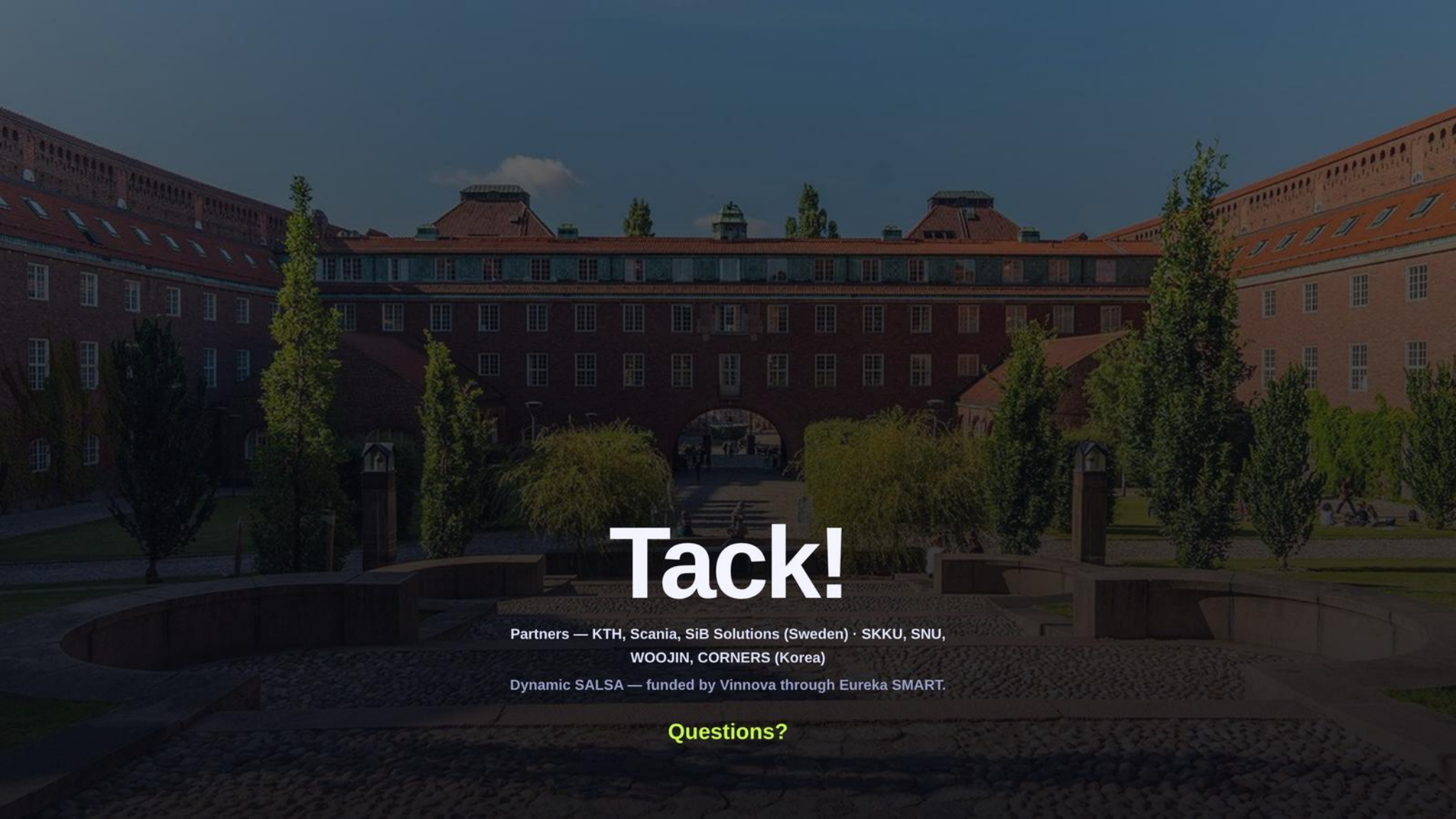
Make logistics flexible & dynamic.

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You're warmly welcome to visit us in Stockholm.





Tack!

Partners — KTH, Scania, SiB Solutions (Sweden) · SKKU, SNU, WOOJIN, CORNERS (Korea)

Dynamic SALSA — funded by Vinnova through Eureka SMART.

Questions?